Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 14)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the energy used per passenger-kilometre by different forms of transport when carrying a typical load.

Table 1

Mode of transport	Typical occupancy	Energy use / MJ per passenger-km
Car (petrol, 1 occupant)	1	2.9
Car (petrol, 4 occupants)	4	0.7
Bus (diesel)	25	0.6
Electric train	200	0.3
Bicycle	1	0.06
Short-haul aircraft	120	2.4

0 1 . 1 Calculate the percentage reduction in energy per passenger-km achieved by carrying four people in a car instead of one.

[1 mark]

Answer ..... %

0 1 . 2 Explain why bulk transport (buses, trains) uses less energy per passenger-km than cars.

[2 marks]

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0 1 . 3 Suggest **three** reasons why people continue to use cars despite the energy advantages of public transport.

[3 marks]

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0 1 . 4 Explain how integrated transport systems and Active Traffic Management reduce energy use.

[4 marks]

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0 2 Table 1 shows three specialist solid wastes.

0 2 . 1 Complete Table 1 to show the main hazard of each waste and the disposal method used.

[6 marks]

Table 1

Waste	Main hazard	Disposal method	
Asbestos			
Cyanide compounds			
High-level radioactive waste			

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0 2 . 2 Explain why encapsulated asbestos in cement roof panels is not considered dangerous while it is intact.

[1 mark]

0 2 . 3 Describe the process of encapsulation used for intermediate-level radioactive waste and wastes containing heavy metals.

[2 marks]

0 2 . 4 Explain why vitrification is preferred to encapsulation in cement for high-level waste.

[1 mark]

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0 3 Table 1 compares a 1000 MW coal-fired power station with a 1000 MW nuclear power station.

Table 1

Feature	Coal-fired	Nuclear (uranium)
Fuel used per year	3.5 million tonnes	30 tonnes
CO_2 emitted per kWh / g	900	12 (mostly fuel processing and construction)
Construction cost / £ billion	2	20
Fuel cost per kWh / pence	3.0	0.5
Typical response to changes in demand	Hours	Slow (days)

0 3 . 1 Explain how nuclear fission releases energy in a reactor.

[3 marks]

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0 3 . 2 Use Table 1 to explain why nuclear power is used for base-load electricity but not for meeting peaks in demand.

[2 marks]

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0 3 . 3 Use Table 1 to explain why nuclear power stations can be built where coal-fired stations could not.

[2 marks]

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0 3 . 4 Explain why the CO_2 emissions of nuclear power are not zero.

[1 mark]

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0 3 . 5 Give **two** reasons why the true cost of nuclear electricity is uncertain.

[2 marks]

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0 4 Table 1 shows some radioactive isotopes, their half-lives and the type of radiation they emit.

Table 1

Isotope	Half-life	Main radiation	Source
Iodine-131	8 days	Beta / gamma	Fission in reactors
Strontium-90	29 years	Beta	Fission in reactors
Caesium-137	30 years	Beta / gamma	Fission in reactors
Plutonium-239	24 400 years	Alpha	Neutron bombardment of U-238
Uranium-238	4.5×10^9 years	Alpha	Occurs naturally in rocks

0 4 . 1 Define the term **half-life**.

[1 mark]

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0 4 . 2 A sample of iodine-131 has an activity of 6400 Bq. Calculate its activity after 32 days.

[2 marks]

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Answer _____ Bq

0 4 . 3 Use Table 1 to explain why isotopes with half-lives of several decades, such as caesium-137, cause the greatest concern after a nuclear accident.

[3 marks]

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0 4 . 4 Explain why plutonium-239 is relatively safe outside the body but very dangerous if inhaled or ingested.

[3 marks]

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0 4 . 5 Explain what is meant by an **activation product**.

[1 mark]

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0 5 Table 1 shows some properties of three pollutants.

Table 1

Pollutant	State at 20 °C	Solubility in water	Solubility in lipids	Environmental half-life
DDT (insecticide)	Solid	Very low	High	About 15 years in soil
Nitrate ions	Dissolved ion	Very high	Very low	Not applicable (not degraded)
Hydrogen cyanide	Gas (denser than air)	High	Low	Hours

0 5 . 1 Define the term **environmental half-life**.

[1 mark]

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0 5 . 2 Use Table 1 to explain why DDT is more likely than nitrate ions to biomagnify in a food chain.

[3 marks]

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0 5 . 3 Suggest why the release of hydrogen cyanide at Bhopal, India, in 1984 caused deaths very close to the source rather than over a wide area.

[2 marks]

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0 5 . 4 Explain how the high water solubility of nitrate ions affects the area over which they cause pollution and the concentration of the pollution.

[2 marks]

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0 5 . 5 Give **two** examples of pollutants that are forms of energy rather than substances.

[2 marks]

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0 6 . 1 Explain why abstraction from an aquifer containing 'fossil' (palaeo) water, such as the High Plains / Ogallala Aquifer in the USA, cannot be sustainable.

[3 marks]

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0 6 . 2 Describe how the GRACE satellites are used to monitor aquifer depletion and give **one** advantage over monitoring boreholes.

[3 marks]

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0 6 . 3 Describe **three** measures being introduced on the North China Plain to reduce aquifer depletion.

[3 marks]

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0 6 . 4 Suggest why large-scale afforestation on the North China Plain may reduce aquifer recharge even though trees reduce soil erosion.

[1 mark]

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0 7 . 1 Explain why it is difficult to be certain that a particular flood or heatwave is caused by anthropogenic climate change.

[3 marks]

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0 7 . 2 Explain why historic temperature records collected in towns may overestimate global warming.

[2 marks]

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0 7 . 3 Describe how each of the following sources of proxy data is used to estimate past climates and give one limitation.

[5 marks]

Tree rings (dendrochronology)

Ice cores

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0 8 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 8 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

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Answer

0 8 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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0 9 A nuclear reprocessing plant on the coast discharges low-level liquid radioactive waste into the sea through a pipeline.

0 9 . 1 Explain what is meant by **Critical Pathway Analysis** (CPA).

[2 marks]

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0 9 . 2 Give **three** factors that would be considered in a CPA for this discharge.

[3 marks]

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0 9 . 3 Describe how **Critical Group Monitoring** (CGM) would be used to assess the risk to the public from this discharge.

[3 marks]

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0 9 . 4 Suggest **two** types of environmental sample that would be collected to check the CPA predictions.

[2 marks]

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1 0 . 1 Explain how organic wastes and inorganic nutrients cause water pollution and evaluate the effectiveness of the methods used to control them.

[9 marks]

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Extra space

1 0 . 2 Give **one** reason why algae cause eutrophication problems more quickly than rooted plants when nutrients are added to a lake.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2 To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS